QUASI-COMPACT OPENS OF QCQS SPACES CUT OUT BY FINITE TYPE IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3396-3431, following the statement at 3382-3394.

Editorial note. No omissions; the surjectivity of |phi^{-1}U| -> |U| is justified in the text.

Allowed standard prerequisites (editorial).

- Affine etale cover of a qcqs space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover (cross-chapter, cited at ~3397-3399). Surjective etale phi : W -> X.
- Quasi-compactness permanence for qcqs sources. Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-quasi-separated-permanence (cross-chapter, cited at ~3403-3405). phi is quasi-compact since X is quasi-separated.
- Scheme case of the ideal criterion. Properties, Lemma properties-lemma-quasi-coherent-finite-type-ideals (cross-chapter, cited at ~3406-3407 and ~3426-3429). Equivalence for the affine scheme W, used in both directions.
- Quasi-compact is local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-local (cross-chapter, cited at ~3408-3409). Lifts quasi-compactness back to U -> X.
- Reduced induced closed subspace ideal. Properties of Spaces, Lemma spaces-properties-lemma-reduced-closed-subspace (cross-chapter, cited at ~3413-3417). The unique quasi-coherent ideal J cutting out the complement.
- Approximating sections by finite type subsheaves. lemma-directed-colimit-finite-type (same chapter, cited at ~3420-3424). Finite type quasi-coherent sub-ideal I agreeing with J on U.

Source reference. Original lemma 0829: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2519–2519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PH: focused-seeds/geometry/S9-spaces-limits-quasi-coherent-finite-type-ideals/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03KG: focused-seeds/geometry/S9-spaces-limits-quasi-coherent-finite-type-ideals/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 073B: focused-seeds/geometry/S9-spaces-limits-quasi-coherent-finite-type-ideals/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H6: focused-seeds/geometry/S9-spaces-limits-quasi-coherent-finite-type-ideals/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IQ: focused-seeds/geometry/S9-spaces-limits-quasi-coherent-finite-type-ideals/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space. Let $U \subset X$ be an open subspace. The following are equivalent:*

1. *$U \to X$ is quasi-compact,*
2. *U is quasi-compact, and*
3. *there exists a finite type quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ such that $|X| \setminus |U| = |V(\mathcal{I})|$.*

Proof. Let W be an affine scheme and let $\varphi : W \to X$ be a surjective étale morphism, see Properties of Spaces, Lemma 03H6. If (1) holds, then $\varphi^{-1}(U) \to W$ is quasi-compact, hence $\varphi^{-1}(U)$ is quasi-compact, hence U is quasi-compact (as $|\varphi^{-1}(U)| \to |U|$ is surjective). If (2) holds, then $\varphi^{-1}(U)$ is quasi-compact because φ is quasi-compact since X is quasi-separated (Morphisms of Spaces, Lemma 073B). Hence $\varphi^{-1}(U) \to W$ is a quasi-compact morphism of schemes by Properties, Lemma 01PH. It follows that $U \to X$ is quasi-compact by Morphisms of Spaces, Lemma 03KG. Thus (1) and (2) are equivalent.

Assume (1) and (2). By Properties of Spaces, Lemma 03IQ there exists a unique quasi-coherent sheaf of ideals $\mathcal{J}$ cutting out the reduced induced closed subspace structure on $|X| \setminus |U|$. Note that $\mathcal{J}|_U = \mathcal{O}_U$ which is an $\mathcal{O}_U$-modules of finite type. As U is quasi-compact it follows from Lemma 0829 that there exists a quasi-coherent subsheaf $\mathcal{I} \subset \mathcal{J}$ which is of finite type and has the property that $\mathcal{I}|_U = \mathcal{J}|_U$. Then $|X| \setminus |U| = |V(\mathcal{I})|$ and we obtain (3). Conversely, if $\mathcal{I}$ is as in (3), then $\varphi^{-1}(U) \subset W$ is a quasi-compact open by the lemma for schemes (Properties, Lemma 01PH) applied to $\varphi^{-1}\mathcal{I}$ on W. Thus (2) holds. □